

Twenty-three Asilomar principles for Artificial Intelligence and the Future of Life¹

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Abstract: The “Asilomar, California Conference on Beneficial AI” was a conference organized by the Institute for the Future of Life (Future of Life Institute) in January 2017, where over a hundred experts and researchers came together to discuss and formulate principles for ethical AI. The twenty-three principles are divided into themes or research questions (five); issues concerning ethics and values (five) and long-term problems (five). This chapter addresses each of the principles, makes them explicit and, finally, analyzes and discusses their feasibility and validity. It is concluded that the movement has received both acclaim and criticism. While some have hailed the principles as a valuable reference point for discussions of AI ethics, others have expressed concern that they lack specific guidance on how to apply them in practice. The question arises as to whether these principles are sufficient to address the complex ethical challenges that arise with artificial intelligence. The answer is that while they might not be enough on their own, they are necessary to initiate and shape this debate.

The Asilomar principles of AI

The so-called “Asilomar AI Principles” [1] were developed as part of a conference that took place in Asilomar, California, in January 2017 and was organized by the Future of Life Institute. (Future of Life Institute) a philanthropic organization dedicated to developing initiatives to solve the world's most pressing problems,

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including artificial intelligence: “From recommendation algorithms to self-driving cars, AI is changing our lives. As the impact of this technology increases, so do its risks” [2, p. 1], warns the Institute.

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While the conference was not open to the public, attendees included Elon Musk, Francesca Rossi, Nick Bostrom, Peter Norvig, Ray Kurzweil, Sang Yong Lee, Stephen Hawking, Stuart Russell, Yann LeCun, Yoshua Bengio, among other personalities. The conference also produced a 14-video series titled “Beneficial AI 2017” which is on YouTube [3].

The final agreements from the conference include principles that cover a wide range of issues, including transparency, accountability, privacy and security. They also contemplate the responsibility of scientists and technology professionals for the impact of their research and the need to involve a wide range of stakeholders in decision-making about the development and use of AI.

The conference brought together more than 100 experts from around the world from different disciplines to discuss how to ensure that the development and use of AI benefit society and do not pose a risk to humans. At the end of the conference, on January 6, 2017, the participants agreed on a set of ethical and responsibility principles to guide the development and application of AI systems [1].

The 2017 “Asilomar, California Conference on Beneficial AI” followed up on a previous conference held in 2015 in Puerto Rico called “The Future of AI: Opportunities and Challenges” [5]. The 2015 conference was also organized by the Future of Life Institute and focused on discussing the potential risks and benefits of artificial intelligence. The 2017 conference built on the discussions and principles formulated at the previous conference but was updated to formulate twenty-three principles for beneficial AI.

The “Asilomar Principles”, until today, are an important guide for the developer community and have been widely disseminated, discussed and analyzed in academia and in the business world [6].

They are divided into three parts: the first, research (five principles); the second, ethics and values (thirteen principles); and the third, long-term aspects (five principles):

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Research topics

1. Research must be beneficial

“The goal of AI research should not be to create undirected intelligence, but beneficial intelligence” [1, p. 1].

AI researchers should strive to create artificial intelligence systems that have a positive impact on society and the world at large. Rather than simply developing intelligent systems without any specific purpose or direction, researchers must work to ensure that their developments are designed and aimed at improving quality of life, productivity, and overall human well-being.

In addition, ethical considerations must be at the forefront of systems research and development. As AI technology advances, it is essential to address concerns such as data privacy and fairness. Researchers must be vigilant in recognizing and mitigating potential harm, so it is important to create a culture of responsible development where transparency and accountability are valued and upheld. Also, interdisciplinary collaboration is crucial in research, as it allows a full understanding of the impact of AI on society. By prioritizing ethical considerations and collaborating across fields, researchers can develop technologies that not only advance the body of knowledge, but also benefit society in concrete ways.

It is necessary to highlight, in this sense, the work from the academy that is currently carried out, for example, by the Institute of Ethics in AI of the University of Oxford (The Ethics in AI Institute) [7]; the Oxford Internet Institute [8]; the Human-Centered Artificial Intelligence Program at Stanford University (Stanford University Human - Centered Artificial Intelligence) [9] and, in Mexico, the Mexican Society of Artificial Intelligence (SMIA) [10], among many other organizations and institutions of research in the world.

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2. Research must be funded

“Research funding: Investments in AI must be accompanied by funding for research that ensures its beneficial use, including contentious issues in computing, economics, law, ethics, and social studies” [1, p. 2].

Here, in accordance with Asilomar's second principle, four dilemmas arise:

The first refers to the need to make future AI systems robust and reliable, so that they work without failures or vulnerabilities to cyberattacks. This is important to ensure that systems can perform their intended functions without causing damage or sending errors that may have negative consequences.

The second dilemma centers on how economic growth can be achieved through automation, without compromising the resources and purpose of people. This refers to the need to ensure that AI does not completely replace human jobs and that it is implemented in a way that improves people's well-being and quality of life, not puts them out of work.

The third dilemma emphasizes the importance of updating legal systems to address the risks associated with AI, while maintaining

fairness and efficiency in the process. This means that legal systems must be reviewed and adapted to the rapid evolution of AI and its implications in the short, medium and long term.

The last question addresses the issue of ethical and legal values that should guide the development and implementation of AI. This implies establishing a governance framework in accordance with human values [1].

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3. Link science with politics

“Science-policy link: There must be a constructive and healthy exchange between AI researchers and political actors” [1, p. 3].

This principle recognizes the importance of collaboration between the scientific community and government officials in addressing the challenges and opportunities presented by artificial intelligence. It recognizes that AI has the potential to significantly transform society, and that its development and deployment must be guided by ethical and social considerations.

The principle implies that researchers and policy makers should engage in an ongoing dialogue to share information, concerns, and perspectives on the development and use of AI. Researchers can and should inform politicians about the latest advances in technology and its potential applications, as well as the risks and challenges they entail. Policy makers can provide input on the legal, ethical, and social implications of AI and help shape regulatory frameworks and public policies that reflect societal values and interests [11].

Some of the main risks associated with AI are job loss due to automation, social manipulation through algorithms (see glossary) and profiling, fake news, social surveillance through devices, algorithmic bias, social inequalities, weakening of values,

autonomous weapons, and speculative algorithms in the stock markets, among many other threats [44].

4. Generate a research culture

“Research culture: A culture of cooperation, trust and transparency between AI researchers and developers should be fostered” [1, p. 4].

This principle recognizes that the development and deployment of AI is a collaborative effort involving researchers, developers, and other stakeholders. It also implies that they must work together in a spirit of openness, cooperation and mutual respect to advance the development of AI in a responsible and beneficial way.

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Developer and researcher are two related but distinct job roles in the field of artificial intelligence. A developer is responsible for designing, refining, and implementing AI-based systems using programming languages and frameworks. A researcher, on the other hand, focuses on advancing the theoretical and practical aspects of AI by conducting experiments, analyzing new algorithms, and publishing articles [43].

The principle envisions that a research culture that fosters cooperation, trust and transparency can help ensure that AI development is in line with collective values and goals. Researchers and developers can work together to share knowledge, experience, and resources to apply best practices and ethical guidelines.

Furthermore, this principle underlines the importance of transparency in AI development. This means that developers need to be open about their methods, data, and conclusions, and try to engage in dialogue with other stakeholders, including policy makers, civil society organizations, and the general public. Promoting

transparency builds trust and ensures that technological development is responsible [12].

5. Security over competitiveness

“Avoiding [commercial] races: Teams developing artificial intelligence systems must cooperate to avoid lowering safety standards” [1, p. 5].

This principle recognizes that AI development is an overly competitive field and that there may be incentives for developers to prioritize speed and efficiency over security and ethical considerations. The principle implies that you should work to establish and comply with security standards, and avoid taking shortcuts to gain comparative and competitive advantage to get to market before others.

The principle of avoiding a commercial race is to avoid jeopardizing protection issues. Instead, developers must prioritize security and ethical considerations, and work collaboratively to set and meet the highest standards for their users and society at large. This can help ensure that AI does not pose unnecessary risk or harm for the sake of profit.

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In addition, the no-race principle suggests that developers should view security standards not as a burden or an obstacle to innovation, but as an essential component of responsible creativity. By working collaboratively to set and enforce standards, developers can help build public confidence in AI and ensure that its development and deployment is responsible and beneficial [13].

Ethics and values

6. AI must be safe

“Security: AI systems must be safe and secure throughout their operational life, and verifiably where appropriate and feasible” [1, p. 6].

This principle recognizes that AI systems have the potential to pose risk and harm if they are not designed, developed and deployed in a responsible and secure manner.

The principle implies that security must be paramount throughout the entire life cycle, from design and development, to deployment and operation of AI. This means implementing features and mechanisms that ensure that the system is reliable in all contexts and scenarios [13].

It also stresses the importance of verifiability, which means that security claims made by developers must be testable through independent means and even controlled attacks. This can help ensure that AI systems are robust in practice and not just theoretically.

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7. Transparent about his mistakes

“Failure transparency: If an AI system causes damage, it must be possible to determine the cause” [1, p. 7].

This principle recognizes that AI systems are not infallible and can fail and even cause harm. The principle implies that when an AI system fails, it is important to understand why and where it went wrong, to avoid similar incidents in the future, and never to minimize or hide what happened [14].

The principle of transparency in failures implies that developers must be open about how the system works, what data it uses and how it makes decisions, avoiding the so-called black box algorithms because what happens inside is unknown. An algorithm is basically

made up of input, process, and output data. In layman's terms, the input-output relationship of a black-box algorithm is known, but the actual steps the algorithm takes to arrive at the result are not transparent. This lack can make it difficult to understand how you make decisions and why you produce certain results. It can be problematic in situations where the decisions made by the algorithm have a significant impact on individuals or society [15]. Some examples of black box algorithms are neural networks, support vector machines (SVM) and some automated decision trees, among others. SVMs are a type of machine learning algorithm (see glossary) that can be used for classification or regression tasks on data with labels or values and can be applied to problems such as signal analysis, artificial understanding of natural languages and the identification of images and sounds [16].

Also, when a system fails, it is important to have processes in place to understand and document the failure, including what went wrong, why it happened, and what steps can be taken to prevent similar events in the future. In black boxes, this is not possible [14].

Likewise, transparency implies that learning is given priority, instead of blaming or covering up mistakes. By being transparent, developers can help build public trust in the AI and know that if something goes wrong, it will at least be known and, second, figure out how to rectify it.

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8. Transparency in judicial matters

"Judicial transparency: Any participation of an autonomous system in judicial decision-making must provide a satisfactory explanation auditable by a competent human authority" [1, p. 8].

This principle recognizes that AI systems are increasingly being used in some judicial systems for risk assessment, sentence recommendation, and other decision-making processes. The

principle implies that, when an AI system participates in judicial decision-making, it is important to ensure that said decision can be explained in a simple, understandable and verifiable way by a human authority [17]. For example, the Hangzhou Intermediate People's Court in China is using AI to help judges make decisions. The AI system, called Xiao Zhi 3.0, has been used in more than 10,000 cases so far [45].

The principle of judicial transparency underlines the importance of responsibility and explainability in the use of AI in the criminal system. This means that developers and users of AI in court must prioritize providing defendants with clear and understandable explanations of how the system has reached a determination, as it has far-reaching implications. In addition, the principle emphasizes the importance of human supervision and participation in the decision-making process to ensure that they are consistent with legal and ethical standards, and that they can be reviewed at all times [17].

Judicial transparency implies that the AI systems used must be designed and developed with accountability in the event of appeals in mind [14].

9. Responsibility

“Designers and users of advanced systems cannot minimize human responsibility when decisions have been made by AI” [1, p. 9].

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This principle recognizes that the development and deployment of AI has moral implications, and that those who design and build these systems have a responsibility to consider and shape their scope. The principle adds that designers and builders of AI systems should play a proactive role in shaping the moral implications of their own systems, rather than just reacting to them as a passive agent [19].

The principle of responsibility underlines the importance of ethical considerations in the design, development and deployment of AI systems. This means that designers and builders must prioritize consideration of the potential impacts and consequences of their systems, and take steps to ensure that they conform to ethical and moral standards [19] and if they do not, respond diligently for it.

For example, making use of AI-generated code is especially risky if users cannot validate it, either because they do not have sufficient technical knowledge or because the tool discourages users from verifying its output. To mitigate these risks, AI-generated code should be treated in the same way as its human-written counterpart. That means applying the same security and liability policies across the board, whether the programming code comes from a human or an AI model [46].

10. Alignment of values

“Aligned values: Highly autonomous AI systems must be designed so that their goals and behaviors can be aligned with human values throughout their operation” [1, p. 10].

This principle recognizes the potential risks associated with autonomous AI systems, capable of making decisions and taking actions without human intervention. The principle involves ensuring that these systems are designed in a way that aligns with human values, to prevent them from acting in a way that is detrimental or inconsistent with established norms [21].

This means that designers and builders of AI systems must give priority to ensuring that the objectives and behaviors of the systems do not act outside the law, and that they do not do so in detrimental ways in violation of Asimov's first rule: “A machine cannot harm a human being, or through inaction allow a human being to come to harm” [22].

11. Respect for human rights

“Human Values: AI systems should be designed and operated in ways that are compatible with ideals of human dignity, rights, freedoms, and cultural diversity” [1, p. eleven].

This principle implies that AI systems must not only conform to ethical and moral norms, but also to broader values related to dignity, inalienable human rights, freedom and cultural diversity.

The principle of human values underscores the importance of designing and operating AI systems in ways that respect the intrinsic value of the person. This means that system designers and builders must give priority to ensuring that they do not violate human dignity, violate human rights or limit cultural diversity [23].

Companies that use AI software to make decisions about healthcare and medicine, employment, and even criminal justice, for example, must answer how they ensure that programs are not knowingly or unknowingly coded with structural biases. The broader adoption of AI in healthcare, autonomous vehicles, and other industries depends on the framework that determines who, if anyone, ends up being held accountable for a violation of people's basic rights by AI systems [24].

12. Privacy of information

“Personal privacy: People should have the right to access, manage and control the data they generate, given the power of AI systems, to analyze and use that data” [1, p. 12].

This principle recognizes the increasing importance of personal data in the AI era, and the potential risks associated with the misuse or inappropriate handling of personal information. It implies that people must have control of the data they generate and the right to

delete it, as well as access it in a way that is consistent with their own interests and preferences.

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The list of personal data includes: first and last name, spouse, children, family, physical address and email address, telephone, date of birth, gender, nationality, passport or official identification number, financial information, credit cards or details of bank accounts, social security number, RFC or any registration as a taxpayer, SSN, CURP, DNI or similar number in each country, medical information, biometrics, employment, personnel number, education, qualifications, IP address, geolocation, usernames, among others.

The privacy principle underlines the importance of protecting personal data and not sharing information that is not the concern of companies, individuals and even governments. This means that AI system designers and builders must prioritize designing systems that are transparent and accountable with respect to the management and use of data, and that allow individuals to exercise control over their own information.

In addition, the principle of personal privacy implies that this data must be protected against unauthorized access or misuse. Unscrupulous individuals are known to sell large databases on the personal data black market [25]. Therefore, priority must be given to designing systems that respect and protect personal privacy and that allow individuals to exercise control over their data [26].

13. Compatibility between privacy and freedom

“Freedom and privacy: The application of AI to personal data must not unreasonably restrict the real or perceived freedom of individuals” [1, p. 13].

This principle recognizes the potential risks associated with the use of AI technologies to analyze and process personal data, and the need to ensure that such use does not lead to unjustified infringements of individual freedoms. The principle highlights the importance of balancing the benefits of AI technology with the need to protect individual liberties and privacy [27].

The principle of freedom and privacy implies that the application of AI to personal data should be subject to appropriate regulatory measures, in order to ensure that individuals are not unduly restricted in their ability to exercise their rights and freedoms. For example, AI systems should not be used to inadvertently collect biometric data from people, as this is considered a violation of their privacy, that is, face recognition, voice recording, iris scanning, or any other biometric data.

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14. Benefits for all

“Co-benefits: AI technologies should benefit and empower as many people as possible” [1, p. 14].

The principle of shared benefit implies that AI technologies must not only be developed and deployed for the benefit of a privileged few, but must also be accessible and available to all. This means that AI technologies must be designed with the needs and interests of various communities in mind, and that the benefits of these technologies must be distributed fairly and equitably, considering all regions of the planet, especially Africa and Latin America.

In addition, the principle of shared benefit underlines the importance of empowering individuals and communities to use AI technologies to achieve their goals and aspirations.

15. Share the prosperity

“Shared Prosperity: The economic prosperity created by AI must be shared widely, for the benefit of all humanity” [1, p. fifteen].

The principle of shared prosperity implies that AI technologies should not only benefit the few who control or own them, but all members of society. It requires that the profits and economic benefits generated by AI technologies be shared, in a way that benefits a broad sector.

The economic benefits of AI should not be concentrated in the hands of a few individuals or companies, but should be distributed more equitably throughout society. This could be achieved through policies and programs that promote greater equality of income and access to economic opportunities, as well as through initiatives that support the development and deployment of AI technologies in ways that are democratized [28].

Some AI systems are powered by millions of low-paid workers around the world, performing repetitive tasks in precarious working conditions, far from Silicon Valley. These exploited workers are often recruited from cheap labor countries such as Kenya, India, the Philippines, and even Mexico [47].

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16. Human control

"Human beings must decide whether or not to delegate their decisions to artificial intelligence systems to achieve the objectives they have chosen" [1, p. 16].

The principle of human control recognizes that AI technologies can automate many tasks and decision-making processes previously performed only by humans. However, it also emphasizes that the ability to decide when and how to delegate decision making to those systems must be retained. This implies that developers should have

the ultimate responsibility and authority to make decisions and set goals for their systems, based on default values [29] [30].

In addition, the human control principle also suggests that one should have the ability to override or intervene in automated decision-making processes, if necessary. This can be especially important in situations where AI may have significant ethical or moral implications.

The principle of human control also underlines the importance of transparency in the decision-making processes of AI systems. This means that humans must be able to understand how systems arrive at their decisions and must have access to information about the algorithms and data used to inform those decisions. The risk of so-called black box algorithms was mentioned earlier [31].

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17. Avoid disruption

“Not subversion: The power conferred by control of highly advanced AI systems should respect and enhance, rather than disrupt, the social and civic processes on which the health of society depends” [1, p. 17].

The principle of non-subversion refers to preventing AI systems from being used to manipulate or undermine democratic processes or individual rights and freedoms. This includes ensuring that AI systems are not used to spread disinformation or manipulate public opinion, as well as ensuring that they do not violate privacy rights or enable new forms of discrimination. There are notorious cases in which public opinion has been manipulated through the generation of messages from bots on social networks to modify political preferences, incite xenophobia or influence their beliefs [32]. A bot – which is an abbreviation for the word robot – is a computer program that performs automated, repetitive, and predefined tasks, often mimicking or replacing the behavior of human users.

Additionally, the non-subversion principle recognizes the importance of ensuring that AI systems are developed in a transparent and accountable manner. For this, AI must be subject to supervision and regulation, and its development must be guided by ethical principles and the common good.

18. AI Armistice

“AI Arms Race: An Arms Race of Lethal Autonomous Weapons Must Be Avoided” [1, p. 18].

The Asilomar principle warns of the imminent dangers of the invention, production and use of lethal autonomous weapons (Lethal Autonomous Weapons or LAWs) [1] [48] and refers to concerns that this could lead to an arms race in which countries or other entities compete to develop and deploy increasingly advanced AI systems that make decisions about the life. Such competition could lead to a dangerous escalation of conflicts and the proliferation of deadly weapons not under human control.

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This principle underscores the importance of ensuring that the development of AI technologies is guided by a commitment to ethical and humanitarian values, and that policy makers take steps to prevent the development of autonomous weapons that could cause harm to civilians or exacerbate conflict. Especially that the Geneva Conventions of 1949, aimed at avoiding barbarism in armed conflicts, are not violated and that crimes against humanity are not committed [33].

It also underlines the need for international cooperation and coordination to face the challenges posed by the rapid development of AI and its use in military contexts, such as the Russian Federation's military offensive against Ukraine since February 2022 [34] and in other armed conflicts.

19. Restriction on future capabilities

“Capacity Caution: In the absence of consensus, strong assumptions regarding upper limits on future AI capabilities should be avoided” [1, p. 19].

This principle recognizes that there is currently no consensus among experts about what the limits of AI capabilities might be, and that there is a risk of making overly optimistic or pessimistic predictions about the potential of technologies. Some have put the self-generative capabilities of GPT4 (Generative Pre-trained Transformer) (see glossary) before arriving at an artificial general or strong intelligence (AGI) that can do everything a human does or feels.

The principle suggests that, given the uncertainty surrounding the development of AI, one should avoid making any sweeping assumptions about its scope. Instead, a prudent approach should be taken, acknowledging the potential risks and benefits of AI and working to ensure that systems are developed in ways that are consistent with human values and priorities [35].

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20. Importance of the future of the earth

“Magnitude: Advanced AI could represent a profound change in the history of life on Earth, and should be planned and managed with adequate care and resources” [1, p. twenty].

This principle underscores that the development of advanced AI systems should be taken seriously and approached cautiously, acknowledging potential risks and benefits. It suggests that resources should be invested to better understand the potential impacts of advanced AI systems and to develop strategies and policies to ensure they are developed and used responsibly and beneficially. This includes the involvement of a wide range of stakeholders, such as AI

experts, policy makers, ethicists, philosophers, and members of the user public, in the development and governance of these systems [36].

21. Dangers of AI

“Risks: Conflicts posed by AI systems, especially catastrophic or existential risks, should be subject to planning and mitigation efforts commensurate with the expected impact” [1, p. twenty-one].

The principle emphasizes the importance of identifying and mitigating the risks posed by AI systems. The statement acknowledges that AI has the potential to help, but also cause significant harm to humanity. Therefore, proactive planning and mitigation efforts are required to responsibly address these risks. The statement also cautions that the potential risks of AI systems are not fully understood and further research is needed to identify and assess them [37].

As AI technology continues to evolve, it is necessary to remain alert to potential risks and prepare accordingly. The impact of these risks could be enormous, and as such, the resources allocated to contain them must be commensurate.

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22. Recursive Self Improvement

“Recursive Autonomous Improvement: AI systems designed to recursively self-improve or self-replicate in ways that can lead to a rapid increase in quality or quantity, must be subject to strict security and control measures” [1, p. 22].

The principle of recursive self-improvement refers to the ability of some AI systems to learn and improve themselves over time; this is known as deep learning or deep learning [38] (see glossary). Recursive self-improvement allows an AI system to continually learn

and make decisions at an ever faster rate. While this can be a valuable feature for AI systems, it also has the potential to create risks and unintended consequences.

Therefore, these systems must be subject to strict security and control measures, and thus minimize the risk of uncontrolled growth or unforeseen consequences derived from automatic improvement. The AI system must be carefully designed and monitored to ensure that it works within established security parameters and does not pose a threat to human safety or well-being. This has given rise to many fictional series [39], but it is also a reality that full autonomy can get out of people's control with unwanted effects. Thus caution, control, and careful planning are required when deploying AI systems that have the potential for recursive self-improvement [40].

23. AI for the common good

“Common good: Superintelligence should only be developed in the service of widely shared ethical ideals, and for the benefit of all humanity and not of a State or organization” [1, p. 23].

This principle is related to the development of artificial general intelligence (AGI), which is, as already stated, an intelligence that has the capacity to perform any intellectual task that a human being can do [41]. Artificial superintelligence (ASI), on the other hand, would be capable of surpassing any human being in any discipline in terms of cognitive abilities, abilities, skills, competencies, etc. Both intelligences, so far, remain only a theoretical possibility.

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For this reason, the principle emphasizes that the development of superintelligence, if it is achieved, should not be driven by individual interests, but should serve the common good and ethical values widely shared among different societies. It implies that the development of AGI should be guided by a worldwide consensus on the ethical principles that should govern its development and use.

On the other hand, as long as there is no regulatory framework that sanctions them, they can represent a danger to humanity. During the World Forum Government Summit in Dubai in March 2023, Elon Musk stated: “One of the biggest risks to the future of civilization is AI, it is positive or negative and it has great, great promise, great capacity... [but also] great danger” [42].

This is why this principle is important because the development of superintelligence has the potential to significantly affect the world and could have far-reaching consequences for everyone. If the AGI or the ASI were to exist, they should be at the service of universal ethical ideals.

Partial conclusions

So far, the twenty-three principles formulated at the end of the Asilomar Conference that, it should be noted, some have been overcome by the invisible hand of the market and the fierce hegemonic-political competition for control of AI. That is, an autonomous warfare in full swing, lack of transparency of many algorithms due to industrial secrecy and racial or class biases, especially in judicial, labor and public security matters, are visible.

However, the Asilomar principles are a set of proposed guidelines to regulate the development and use of AI in a responsible manner. The 2017 “Asilomar Principles for AI” aim to ensure that it is developed safely and that it benefits society.

While these principles are an important starting point for addressing the ethical and security issues raised by AI, it is important to note that they will not solve all of the problems and challenges associated with AI. Ongoing discussions, stakeholder exchanges, regulations, and concerted efforts are needed to ensure that AI is developed and used responsibly.

One criticism of Asilomar's twenty-three principles and the broader AI ethics movement is that they are too general and lack specific guidance on how to apply them in practice. Some like Garbowski [6] question whether they are sufficient to address the complex ethical challenges posed by AI, but it could be answered that they are; Although they are not sufficient, they are necessary and represent a foundational framework that must be further developed and refined to provide more specific guidance on the practical application of ethical principles. In general, the twenty-three principles have been well received by the community and are often cited as a key reference point in formal and informal discussions of AI ethics.

Referencias

- [1] Future of Life Institute, "The Asilomar AI Principles," Futureoflife.org Acceso ene. 2023. [En línea] Disponible: <https://futureoflife.org/open-letter/ai-principles/>
- [2] Future of Life Institute, "Cause Area Artificial Intelligence," Futureoflife.org Acceso ene. 2023. [En línea] Disponible: <https://futureoflife.org/cause-area/artificial-intelligence/>
- [3] Future of Life Institute, "Beneficial AI 2017." (30 de enero de 2017). [Video en línea]. Disponible: <https://bsu.buap.mx/b0e>
- [4] Future of Life Institute, "Steering transformative technology towards benefitting life and away from extreme large-scale risks," Futureoflife.org Acceso ene. 2023. [En línea] Disponible: <https://futureoflife.org/>
- [5] Future of Life Institute, "The Future of AI: Opportunities and Challenges," Futureoflife.org Acceso ene. 2023. [En línea] Disponible: <https://futureoflife.org/event/ai-safety-conference-in-puerto-rico/>
- [6] M. Garbowski, "A critical analysis of the Asilomar AI principles," Zeszyty Naukowe, vol. 115, pp. 45-55, 2017, <https://bsu.buap.mx/cjb>

- [7] Oxford Institute for Ethics in AI, "Institute for Ethics in AI." Acceso ene. 2023. [En línea] Disponible: <https://www.oxford-aiethics.ox.ac.uk/>
- [8] Oxford Internet Institute, "Oxford Internet Institute," Acceso ene. 2023. [En línea] Disponible: <https://www.oii.ox.ac.uk/>
- [9] HAI. "Stanford University Human-Centered Artificial Intelligence," Stanford.edu. Acceso ene. 2023. [En línea] Disponible: <https://hai.stanford.edu/>
- [10] SMIA, "Sociedad Mexicana de Inteligencia Artificial," Acceso ene. 2023. [En línea] Disponible: <https://smia.mx/>
- [11] V. Durrer, T. Miller, L. A. Celi, y M. Ghassemi, "The Routledge Handbook of Global Cultural Policy," 1st ed. Abingdon: Routledge, 2018.
- [12] J. Kroll, "Accountability in Computer Systems," en The Oxford Handbook of Ethics of AI, M. Dubber, F. Pasquale, y S. Das, Eds., Oxford University Press, 2020, pp. 180-196.

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- [13] C. Stadlmann y A. Zehetner, "Human Intelligence Versus Artificial Intelligence: A Comparison of Traditional and AI-Based Methods for Prospect Generation," en Marketing and Smart Technologies, Springer, 2021, pp. 11-22.
- [14] N. Diakopoulos, "Transparency. Accountability, Transparency, and Algorithms," en The Oxford Handbook of Ethics of AI, M. Dubber, F. Pasquale, y S. Das, Eds. Oxford: Oxford University Press, 2020, pp. 197-213, doi: 10.1093/oxfordhb/9780190067397.013.11.
- [15] L. Floridi, "Ethics, Governance, and Policies in Artificial Intelligence," Cham: Springer, 2021.
- [16] G. Z. Yang et al., "The grand challenges of Science Robotics," Science Robotics, vol. 3, no. 14, p. eaar7650, ene. 2018, doi: 10.1126/scirobotics.aar7650
- [17] H. Surden, "Ethics of AI in Law: Basic Questions," en The Oxford Handbook of Ethics of AI, M. Dubber, F. Pasquale, y S. Das, Eds., Oxford: Oxford University Press, 2020, pp. 719-736.

- [18] W. Schröder, "Robots and Rights: Reviewing Recent Positions in Legal Philosophy and Ethics," en *Robotics, AI, and Humanity: Science, Ethics, and Policy*, J. von Braun et al., Eds., Springer International Publishing, Cham, 2021, pp. 191-203.
- [19] C. Bartneck, C. Lütge, A. Wagner, y S. Welsh, "Responsibility and Liability in the Case of AI Systems," en *An Introduction to Ethics in Robotics and AI*, C. Bartneck, et al., Eds. Springer International Publishing, 2021, pp. 39-44.
- [20] D. Gunkel, "Perspectives on Ethics of AI: Philosophy," in *The Oxford Handbook of Ethics of AI*, M. Dubber, F. Pasquale, y S. Das, Eds., Oxford University Press, 2020, pp. 538-553.
- [21] A. Korinek, "Integrating Ethical Values and Economic Value to Steer Progress in Artificial Intelligence," en *The Oxford Handbook of Ethics of AI*, M. Dubber, F. Pasquale, y S. Das, Eds., Oxford University Press, 2020, pp. 475-491.
- [22] I. Asimov, "Strange playfellow," *Super Science Stories*, vol. 1, no. 4, pp. 67-77, 1940.
- [23] L. Lim y H. K. Lee, "Routledge handbook of creative and cultural industries in Asia," Routledge handbooks, Abingdon, UK: Routledge, 2019.
- [24] K. Yeung, A. Howes, y G. Pogrebna, "AI Governance by Human Rights-Centered Design, Deliberation, and Oversight: An End to Ethics Washing," en *The Oxford Handbook of Ethics of AI*, M. Dubber, F. Pasquale, y S. Das, Eds., Oxford University Press, 2020, pp. 76-106.
- [25] C. Jordan y D. Maimon, "New research shows that darknet markets net millions selling stolen personal data," *Fastcompany.com*. Acceso ene. 2023. [En línea] Disponible: <https://bsu.buap.mx/b0G>
- [26] J. Antoniou y O. Tringides, "Personal Data, Cloud Platforms, Privacy and Quality of Experience," in *Effects of Data Overload on User Quality of Experience*, J. Antoniou y O. Tringides, Eds., Cham: Springer International Publishing, 2023, pp. 37-54.
- [27] C. Bartneck, C. Lütge, A. Wagner y S. Welsh, "Privacy Issues of AI," in *An Introduction to Ethics in Robotics and AI*, C. Bartneck et al., Eds., Springer, 2021, pp. 61-70.

- [28] H. Dang y P.F. Lanjouw, “Toward a New Definition of Shared Prosperity: A Dynamic Perspective from Three Countries,” en *Inequality and Growth: Patterns and Policy: Volume I: Concepts and Analysis*, K. Basu and J.E. Stiglitz, Eds., Palgrave, 2016, pp. 151-171.
- [29] F. Morandín-Ahuerma, “Leyendas de trolley: juicio moral y toma de decisiones,” *Universita Ciencia*, vol. 8, no. 22, pp. 79-91, 2019.
- [30] J. Morley, L. Floridi, L. Kinsey y A. Elhalal, “From What to How: An Initial Review of Publicly Disponible en AI Ethics Tools, Methods and Research to Translate Principles into Practices,” en *Ethics, Governance, and Policies in Artificial Intelligence*, L. Floridi, Ed., Cham: Springer International Publishing, 2021, pp. 153-183.
- [31] L. Ibarra, D. Balderas, P. Ponce y A. Molina, “Fast Execution of Black-Box Algorithms Through a Piece-Wise Linear Interpolation Technique,” *Arab. J. Sci. Eng.*, vol. 44, no. 11, pp. 9443-9453, 2019, doi: 10.1007/s13369-019-04042-y.
- [32] J. Mökande, J. Morley, M. Taddeo y L. Floridi, “Ethics-Based Auditing of Automated Decision-Making Systems: Nature, Scope, and Limitations,” *Sci. Eng. Ethics*, vol. 27, no. 4, p. 44, 2021, doi: 10.1007/s11948-021-00319-4.
- [33] J. Galliot y J. Scholz, “The Case for Ethical AI in the Military,” en *The Oxford Handbook of Ethics of AI*, M. Dubber, F. Pasquale, y S. Das, Eds., Oxford University Press, 2020, pp. 684-702, doi: 10.1093/oxfordhb/9780190067397.013.43.
- [34] S. Russell, “AI weapons: Russia’s war in Ukraine shows why the world must enact a ban,” *Nature*, vol. 614, no. 7949, pp. 620-623, 2023.
- [35] T. Powers y J. Ganascia, “The Ethics of the Ethics of AI,” in *The Oxford Handbook of Ethics of AI*, M. Dubber, F. Pasquale, y S. Das, Eds., Oxford University Press, 2020, pp. 26-51, doi: 10.1093/oxfordhb/9780190067397.013.2.
- [36] S. Russell y P. Norvig, “Philosophy, ethics, and safety of AI,” en *Artificial Intelligence: A Modern Approach*, Londres: Pearson, 2022, pp. 1032-1062.

- [37] T. Winkle, “Product Development within Artificial Intelligence, Ethics and Legal Risk.” Cham: Springer Vieweg.
- [38] H.P. Cowley et al., “A framework for rigorous evaluation of human performance in human and machine learning comparison studies,” *Scientific Reports*, vol. 12, no. 1, p. 5444, 2022, doi: 10.1038/s41598-022-08078-3.
- [39] E. Saffari, S. R. Hosseini, A. Taheri y A. Meghdari, “Does cinema form the future of robotics? a survey on fictional robots in sci-fi movies,” *SN Applied Sciences*, vol. 3, no. 6, p. 655, 2021, doi: 10.1007/s42452-021-04653-x.
- [40] A. Majot y R. Yampolskiy, “Diminishing Returns and Recursive Self Improving Artificial Intelligence,” en *The Technological Singularity: Managing the Journey*, V. Callaghan et al., Eds., Springer Berlin Heidelberg, 2017, pp. 141-152.
- [41] T. Vasil, P. Skalfist, y D. Mikelsten, “Inteligencia artificial: la cuarta revolución industrial,” Cambridge Stanford Books, 2020.

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- [42] T. Barrabi, “Elon Musk warns AI 'one of biggest risks' to civilization during ChatGPT's rise,” *NYPPost.com*, Acceso ene. 2023. [En línea] Disponible: <https://nypost.com/2023/02/15/elon-musk-warns-ai-one-of-biggest-risks-to-civilization/>
- [43] S. Gupta, “Data Scientist vs. Artificial Intelligence Engineer: Which Is a Better Career Choice?”, Acceso ene. 2023. [En línea] Disponible: <https://bsu.buap.mx/b3P>
- [44] M. Thomas, “8 Risks and Dangers of Artificial Intelligence to Know.” *Builtin.com*. Acceso ene. 2023. [En línea] Disponible: <https://builtin.com/artificial-intelligence/risks-of-artificial-intelligence>.
- [45] DeutscheWelle, “Cortes chinas ya resuelven casos con inteligencia artificial.” *DW.com*. Acceso ene. 2023. [En línea] Disponible: <https://www.dw.com/es/las-cortes-de-china-ya-utilizan-inteligencia-artificial-para-resolver-casos/a-64471873>
- [46] L. Craig, “The promises and risks of AI in software development,” *Techtarget*, Acceso ene. 2023, [En línea]. Disponible:

<https://www.techtarget.com/searchitoperations/feature/The-promises-and-risks-of-AI-in-software-development>

- [47] A. Williams, M. Miceli y T. Gebru, "The Exploited Labor Behind Artificial Intelligence," Acceso ene. 2023. [En línea]. Disponible: <https://www.noemamag.com/the-exploited-labor-behind-artificial-intelligence/>

- [48] M. Taddeo y A. Blanchard, "Accepting Moral Responsibility for the Actions of Autonomous Weapons Systems—a Moral Gambit," *Phil. & Tech.*, vol. 35, no. 3, p. 78, 2022/08/05 2022, doi: 10.1007/s13347-022-00571-x.